

BIT Sindri, Dhanbad

Electrical Engineering Department

Assignment-2

Fundamental of Measurement and Sensor

Unit 3

1. Assess the differential output with reference to LVDT.
2. Sketch the equivalent circuit for LVDT.
3. What is the principle of variable reluctance transducer?
4. Summarize in detail about the characteristics of LVDT and any two applications of LVDT.
5. Analyze the capacitive displacement transducers based on change in distance between plates and also explain why differential arrangement is needed?
6. List out any four materials by which piezoelectric transducers are made of.
7. Illustrate the principle of 'Hall effect' with a sketch.
8. How a piezoelectric transducer could be used as an accelerometer?
9. Define piezoelectric effect. Draw the equivalent circuit of a piezoelectric crystal?
10. Describe the principle of operation of hall transducer for displacement and current measurement.